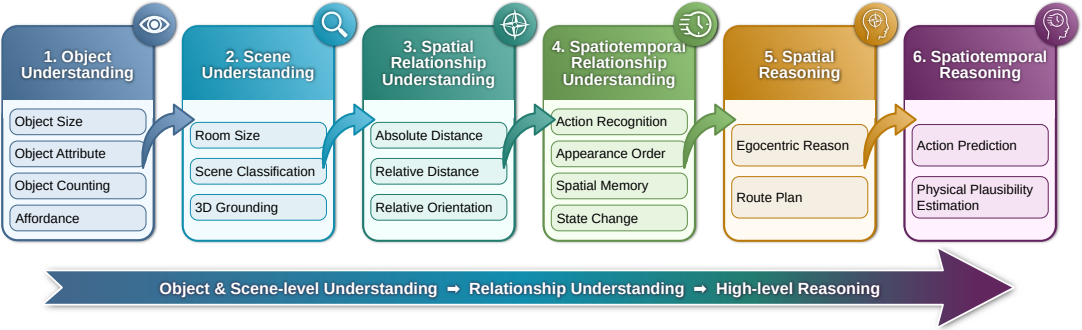
1. Object Understanding
Object Size
Object Attribute
Object Counting
Affordance
2. Scene Understanding
Room Size
Scene Classification
3D Grounding
3. Spatial Relationship Understanding
Absolute Distance
Relative Distance
Relative Orientation
4. Spatiotemporal Relationship Understanding
Action Recognition
Appearance Order
Spatial Memory
State Change
5. Spatial Reasoning
Egocentric Reason
Route Plan
6. Spatiotemporal Reasoning
Action Prediction
Physical Plausibility Estimation
Object & Scene-level Understanding ➡ Relationship Understanding ➡ High-level Reasoning